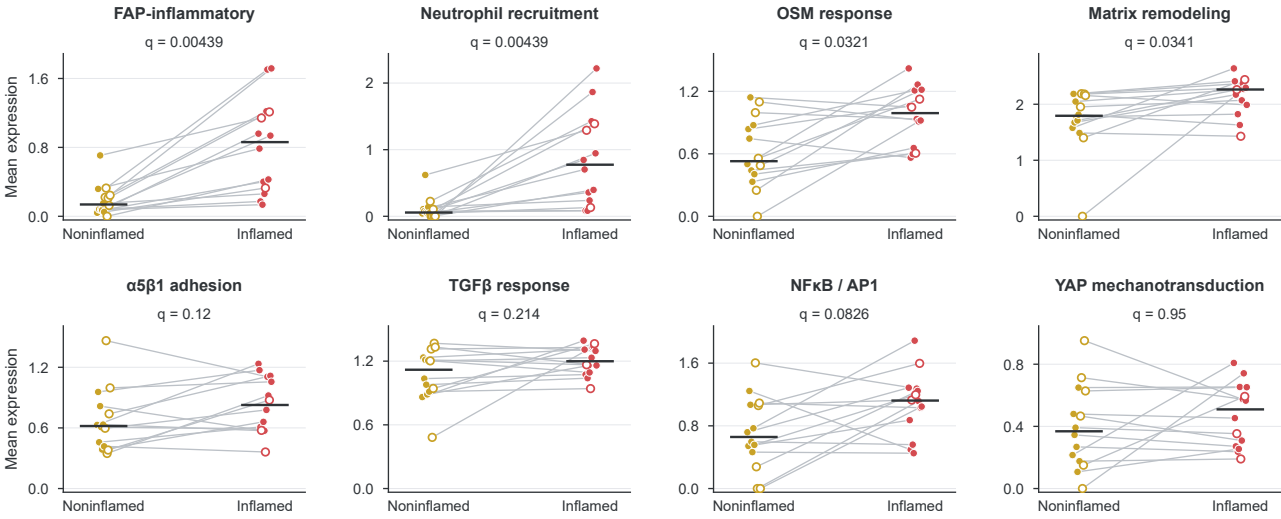


Supplementary Figure 1

a Paired stromal expression of fibroblast-associated programs

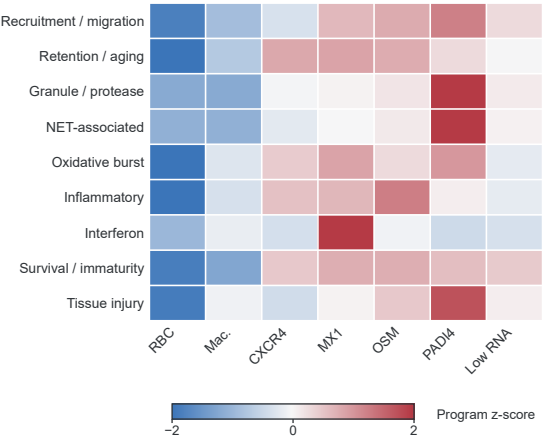
Matched noninflamed and inflamed UC tissue; 14 patients



Paired Wilcoxon q (BH, 8 programs). Lines: patients; bars: medians; open points: <10 recovered cells.

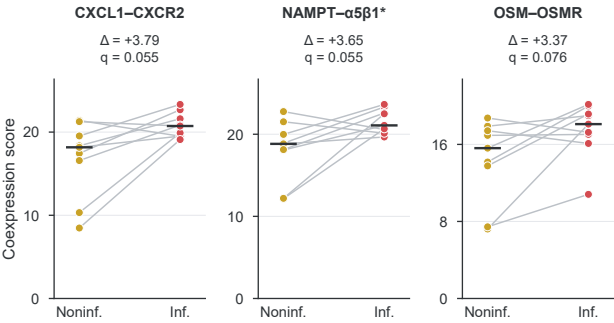
b Pooled RNA programs and annotation controls

All conditions; original gene modules



c Paired ligand–receptor coexpression

9 matched UC patient pairs



Paired Wilcoxon q (BH, 5 original routes).

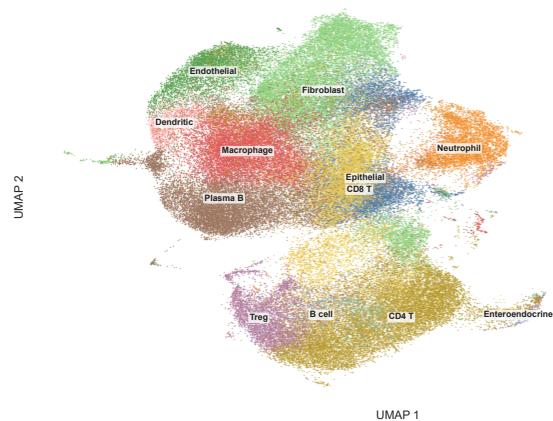
*Receptor summary: min(ITGA5, ITGB1).

Δ : median paired change; lines: patients.

Supplementary Figure 2

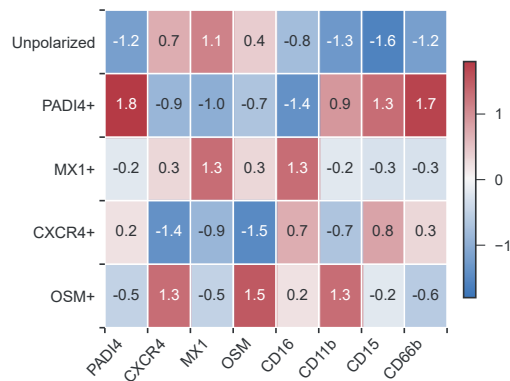
a Broad CODEX cell atlas

245,020 cells; 12 broad annotations



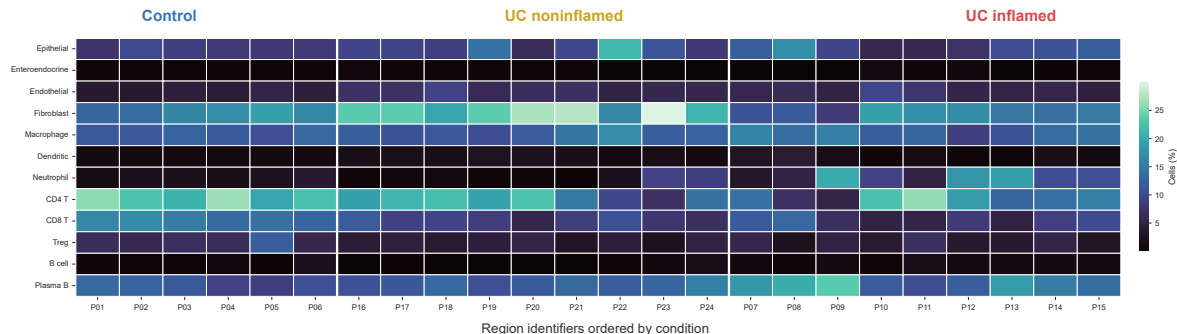
b Neutrophil phenotype validation

Median marker expression, z-scored across states



c Region-resolved cell composition

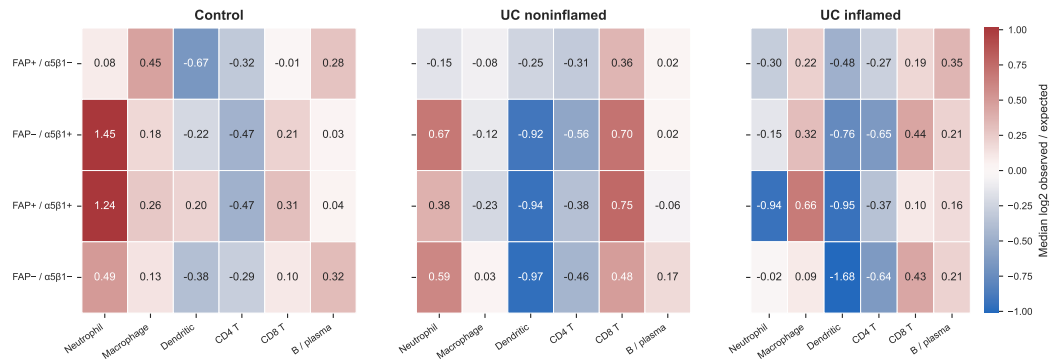
Six control, nine noninflamed UC and nine inflamed UC region identifiers



Supplementary Figure 3

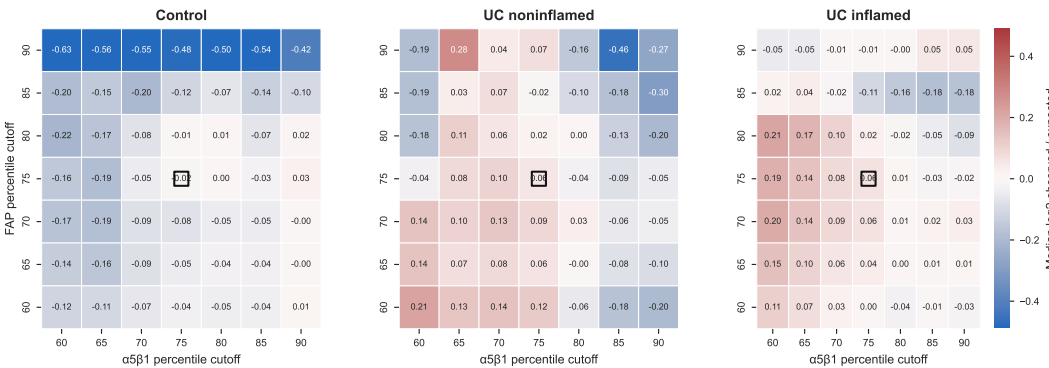
a Immune-lineage spatial specificity

Median log2 observed/expected proximity across FAP/ $\alpha 5\beta 1$ strata



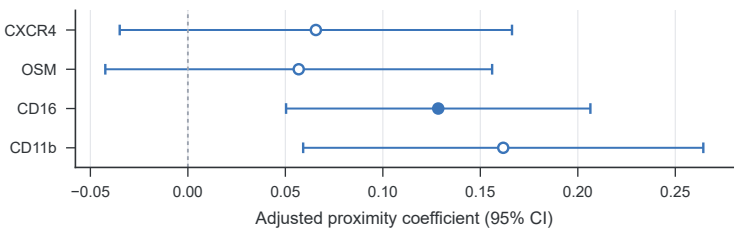
b Threshold sensitivity

FAP and $\alpha 5\beta 1$ percentile cutoffs across tissue conditions



c Adjusted neutrophil proximity effects

Filled point: restricted spatial-null $q < 0.05$; bars show 95% CI

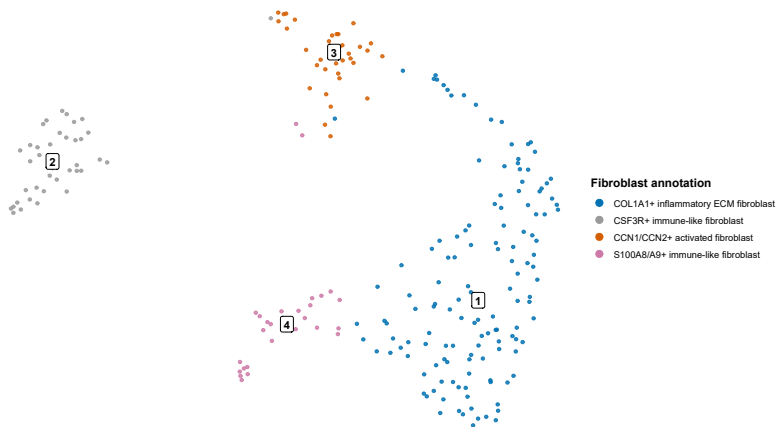


Model q	Spatial-null q
0.242	0.844
0.242	0.843
0.008	0.032
0.008	0.144

Supplementary Figure 4

a Fibroblast subclusters and annotations

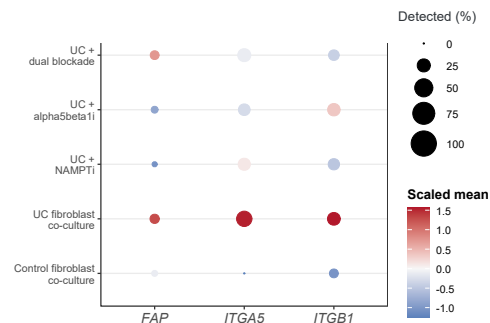
Donor-harmonized fibroblast embedding; annotation resolution 0.4



Immune-like annotations are retained with low confidence because their stromal programs are weak.

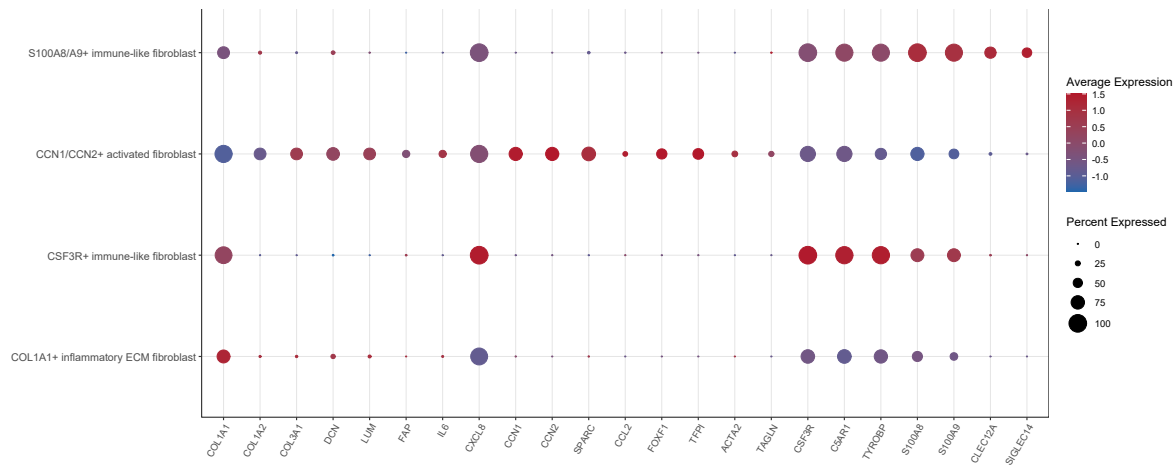
b Target transcripts by condition

FAP, ITGA5 and ITGB1 expression



c Marker programs supporting annotations

Dot size: percent expressed; color: scaled mean expression



Supplementary Figure 5

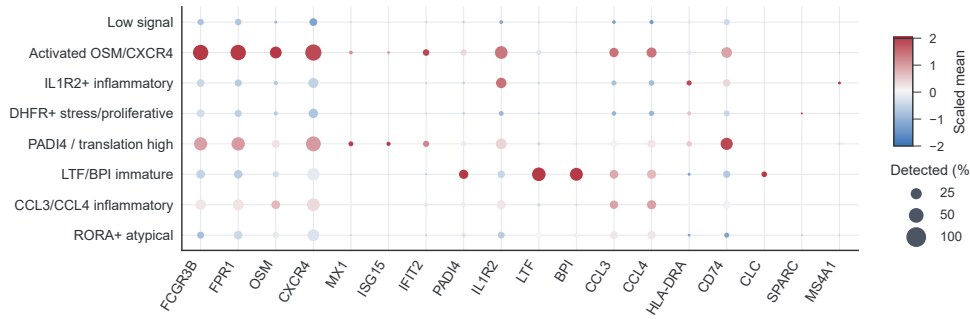
a Neutrophil subcluster atlas

Retained cells colored by transcriptomic subcluster annotation



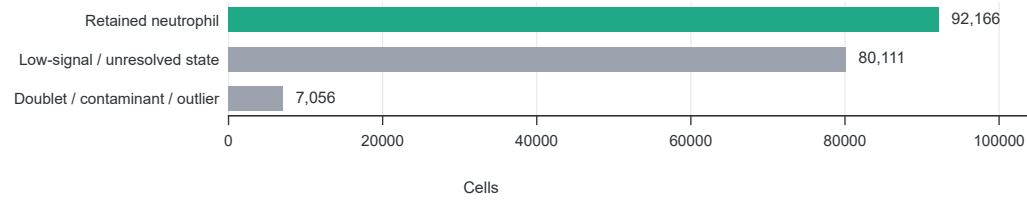
b Marker support for subcluster annotations

Dot size: detected cells; color: scaled mean expression



c Cell-retention and cleanup inventory

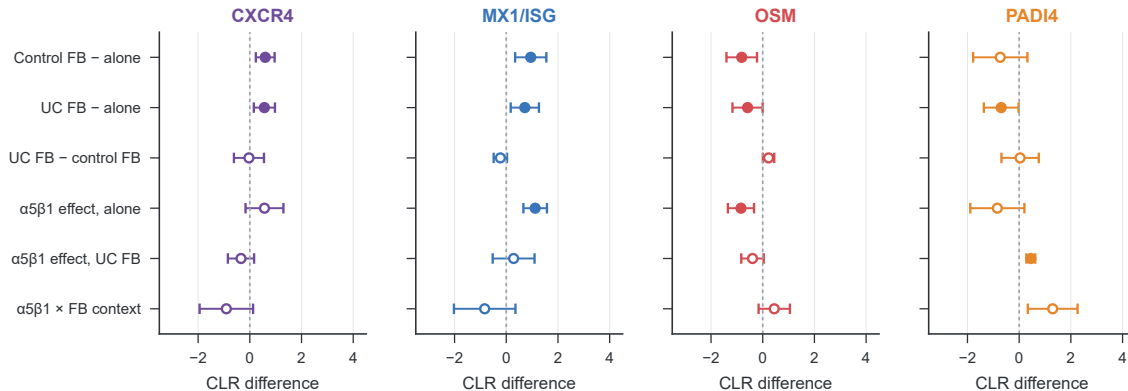
Original counts summarized by disposition



Supplementary Figure 6

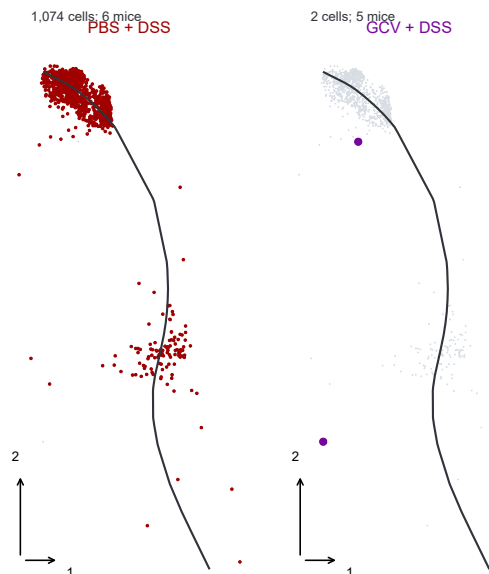
a Donor-aware neutrophil-state contrasts

Six recorded donors; filled points denote BH $q < 0.05$; bars show 95% CI



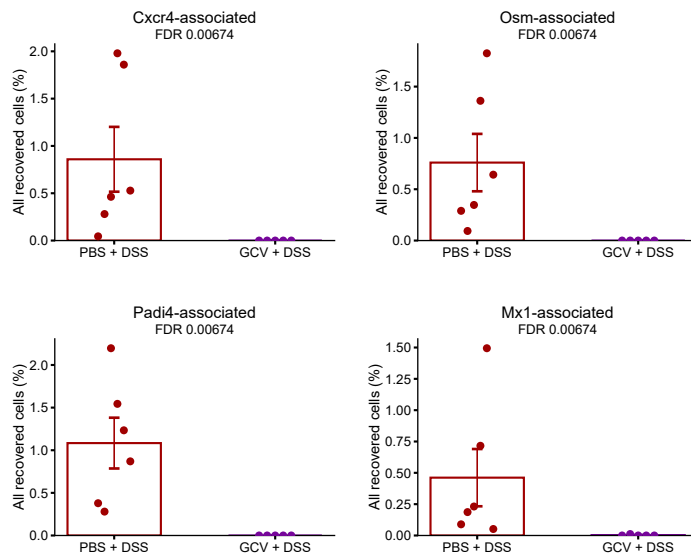
Supplementary Figure 7

a Neutrophil occupancy after FAP ablation

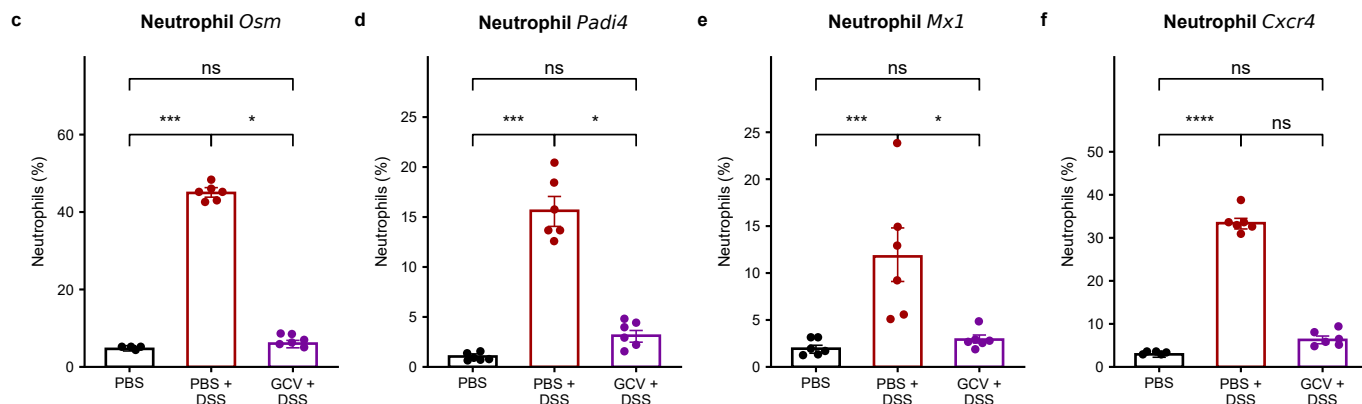


Only two postablation neutrophils: occupancy loss, not a postablation trajectory or within-state estimate.

b Recovered neutrophil-state fractions



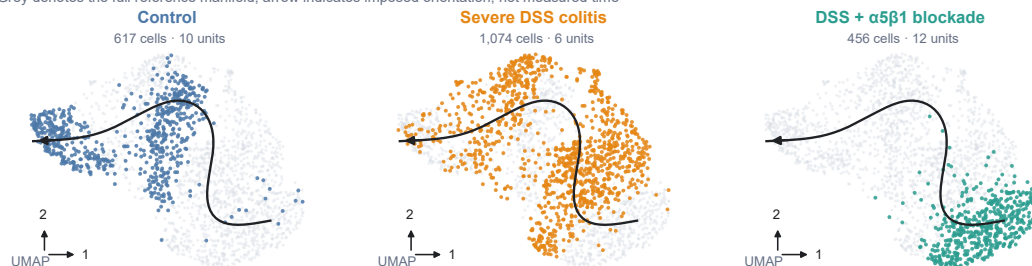
c-f Neutrophil marker readouts supporting Figure 4



Supplementary Figure 8

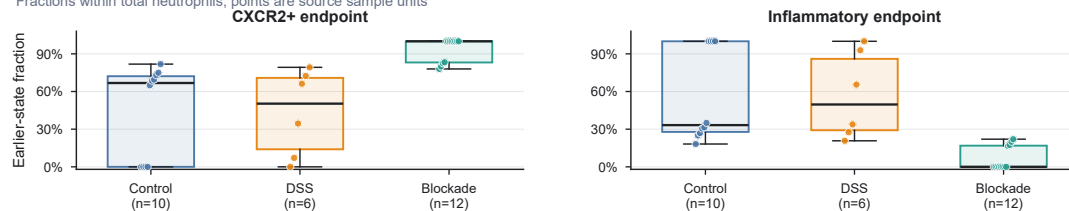
a $\alpha 5\beta 1$ blockade redistributes occupancy along a shared neutrophil continuum

Grey denotes the full reference manifold; arrow indicates imposed orientation, not measured time



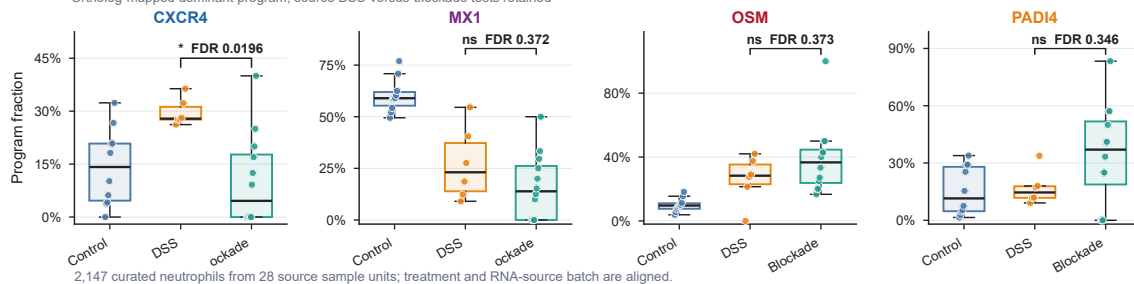
b Endpoint-state redistribution

Fractions within total neutrophils; points are source sample units



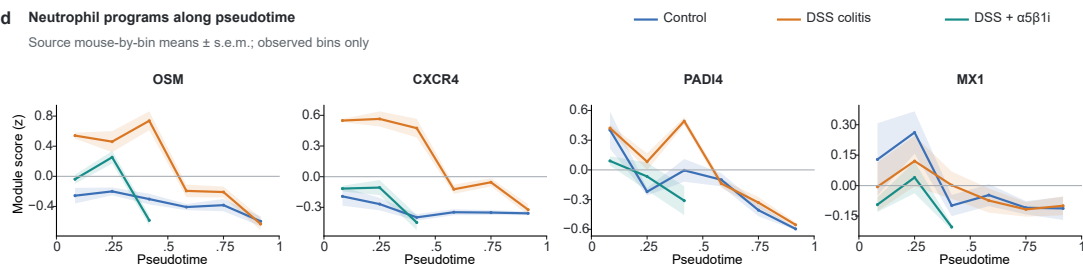
c Transferred human-associated neutrophil programs

Ortholog-mapped dominant program; source DSS-versus-blockade tests retained



d Neutrophil programs along pseudotime

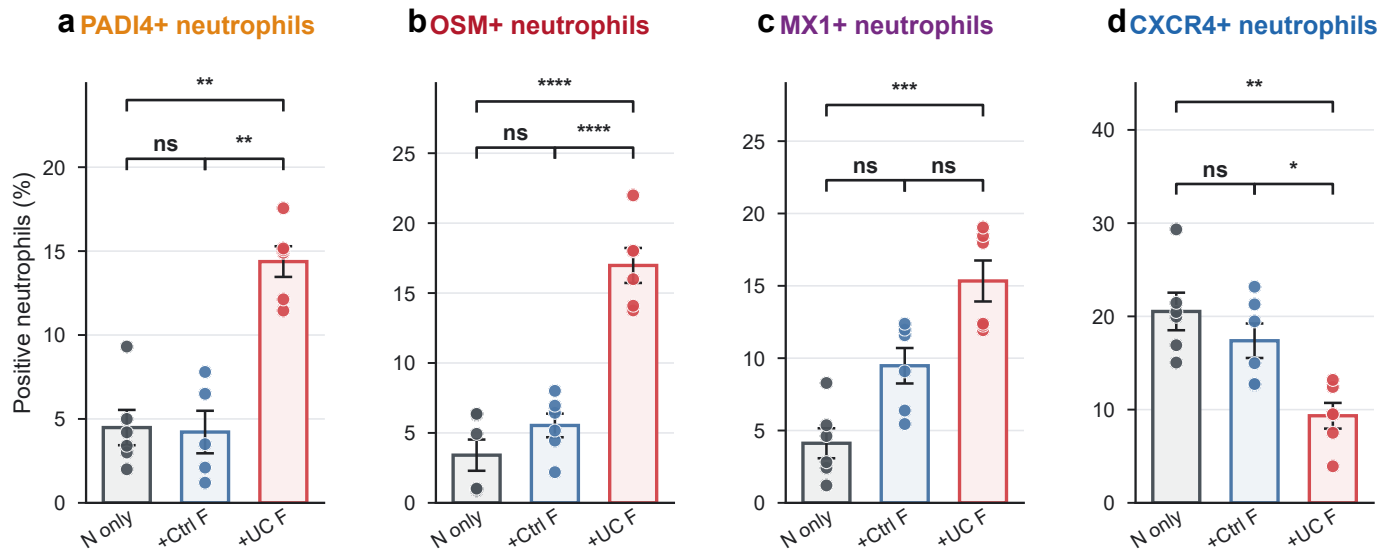
Source mouse-by-bin means $\pm$ s.e.m.; observed bins only



Supplementary Figure 9

Human fibroblast coculture: neutrophil protein readouts

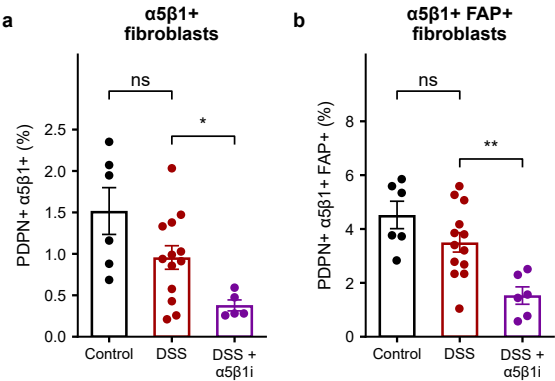
Neutrophils only + control fibroblasts + UC fibroblasts



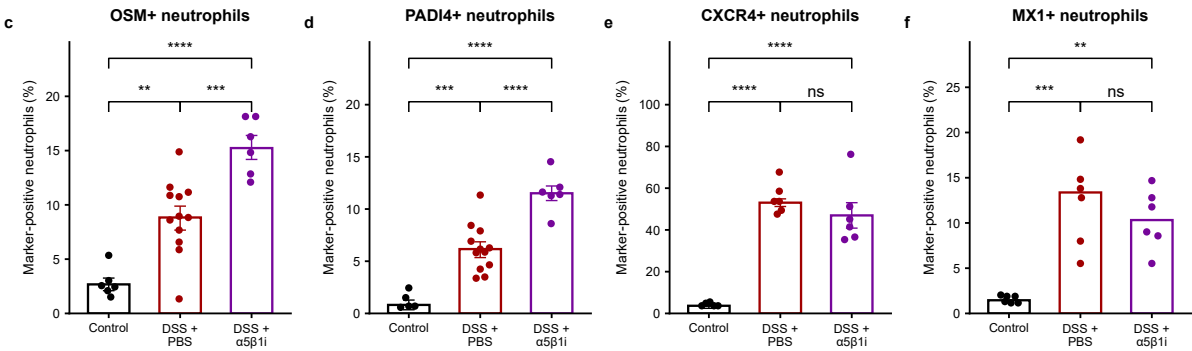
Supplementary Figure 10

DSS colitis: stromal and neutrophil protein readouts after alpha5beta1 blockade

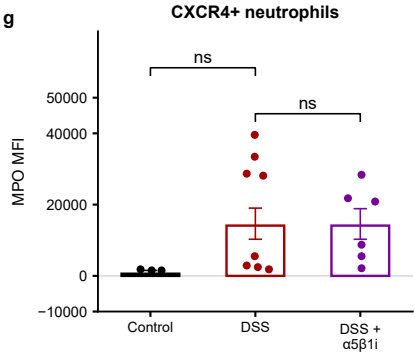
a,b Stromal alpha5beta1 detection



c-f Marker-positive neutrophils



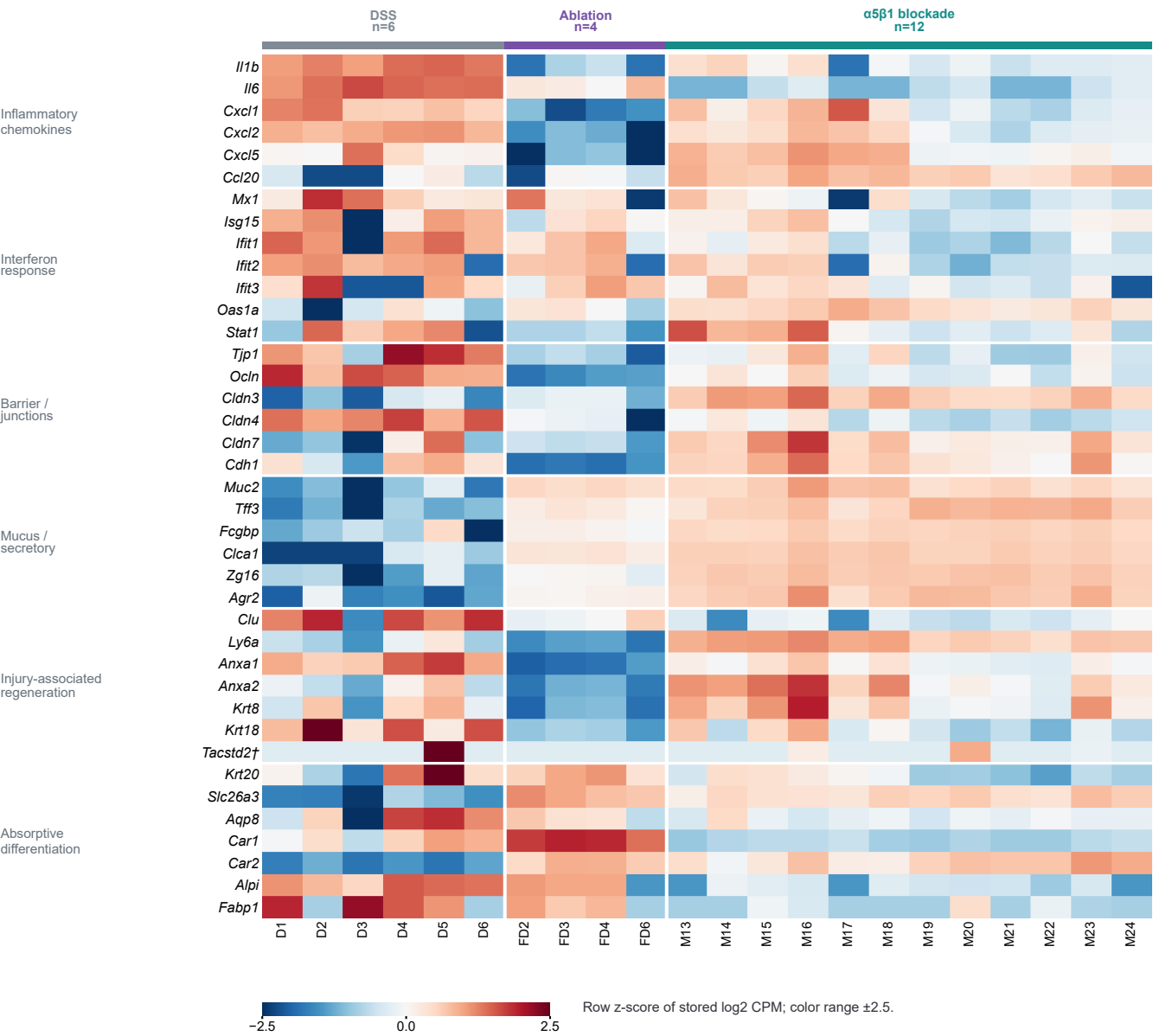
g MPO within marker-positive neutrophils



Supplementary Figure 11

Complete epithelial program gene expression

Mature absorptive colonocytes; each column is one biological mouse



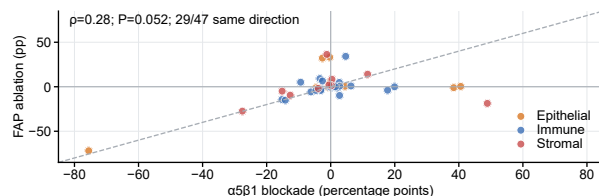
†Tacstd2 detected in 4/22 mice. Descriptive display; no gene-level hypothesis tests.

Supplementary Figure 12

Cell-state concordance and predicted communication

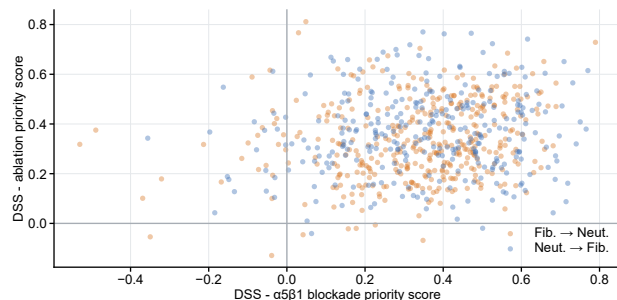
Separate stromal interventions compared with a shared DSS reference

a Cell-state response concordance



b Predicted interaction concordance

691 routes: $\rho=0.15$, $P=0.00013$; core: 490/502 attenuated



Panel c: intervention key

● FAP-lineage ablation

▲ $\alpha 5 \beta 1$ blockade

Positive values indicate lower predicted communication priorities relative to DSS.

c Predicted reciprocal routes

